

Sitagliptin Tablets

Change to read:

DEFINITION

Sitagliptin Tablets contain an amount of sitagliptin phosphate ($C_{16}H_{15}F_6N_5O \cdot H_3PO_4$) (USP 1-Aug-2022) equivalent to NLT 95.0% and NMT 105.0% of the labeled amount of sitagliptin ($C_{16}H_{15}F_6N_5O$).

IDENTIFICATION

Delete the following:

- ▲ **A. SPECTROSCOPIC IDENTIFICATION TESTS** (197), *Infrared Spectroscopy*: 197K or 197A

Sample: Grind a Tablet, and use a suitable amount of the powdered Tablet to prepare a specimen.

Analysis: Record the spectra of the *Sample* and the *Standard* using 197A or 197K, and examine the spectra in the range between 1800 cm^{-1} and 650 cm^{-1} .

Acceptance criteria: The spectrum obtained from the *Sample* shows bands at approximately 1669 , 1513 , 1425 , 1207 , 880 , and 844 cm^{-1} , similar to the spectrum from the *Standard* similarly obtained. [NOTE—Peak positions may vary slightly between instruments (within $\pm 10\text{ cm}^{-1}$). Other peaks may be present in the spectra that do not appear in this list.]▲ (USP 1-Aug-2022)

Add the following:

- ▲ **A.** The UV spectrum of the major peak of the *Sample solution* corresponds to that of the *Standard solution*, as obtained in the *Assay*.▲ (USP 1-Aug-2022)
- **B.** The retention time of the major peak of the *Sample solution* corresponds to that of the *Standard solution*, as obtained in the *Assay*.

ASSAY

Change to read:

• PROCEDURE

Buffer: 1.36 g/L of monobasic potassium phosphate, adjusted with phosphoric acid to a pH of 2.0

Mobile phase: Acetonitrile and *Buffer* (15:85)

Dilute phosphoric acid: Transfer 1 mL of phosphoric acid to a 1-L volumetric flask, and dilute with water to volume.

Diluent: Acetonitrile and *Dilute phosphoric acid* (5:95)

Standard solution: 0.1 mg/mL of USP Sitagliptin Phosphate RS in *Diluent*

Sample stock solution: Nominally 1.0 mg/mL of sitagliptin, prepared as follows. Place 10 Tablets in a suitable volumetric flask, and dilute with *Diluent* to volume. Add a stir bar and stir vigorously for 1 h, ensuring that a vortex is achieved during stirring.

Sample solution: Nominally 0.08 mg/mL of sitagliptin, prepared as follows. Transfer 8.0 mL of the *Sample stock solution* to a 100-mL volumetric flask, and dilute with *Diluent* to volume. Centrifuge a portion of the solution for 10 min or until the solution is clear, and use the supernatant.

Chromatographic system

(See *Chromatography* (621), *System Suitability*.)

Mode: LC

Detector: UV 205 nm. ▲For *Identification A* use a diode array detector in the range of 210–400 nm.▲ (USP 1-Aug-2022)

Column: 4.6-mm $\times$ 15-cm; 5- μ m packing L10

Column temperature: 30°

Flow rate: ▲1 mL/min▲ (USP 1-Aug-2022)

Injection volume: 20 μ L

System suitability

Sample: *Standard solution*

Suitability requirements

Relative standard deviation: NMT 2.0%

Analysis

Samples: *Standard solution* and *Sample solution*

Calculate the percentage of the labeled amount of sitagliptin ($C_{16}H_{15}F_6N_5O$) in the portion of Tablets taken:

$$\text{Result} = (r_U/r_S) \times (C_S/C_U) \times (M_{r1}/M_{r2}) \times 100$$

r_U = peak area from the *Sample solution*

r_S = peak area from the *Standard solution*

C_S = concentration of USP Sitagliptin Phosphate RS in the *Standard solution* (mg/mL)

C_U = nominal concentration of sitagliptin in the *Sample solution* (mg/mL)

M_{r1} = molecular weight of sitagliptin, 407.32

M_{r2} = molecular weight of anhydrous sitagliptin phosphate, 505.31

Acceptance criteria: 95.0%–105.0%

PERFORMANCE TESTS

• DISINTEGRATION (701)

Time: 5 min

Acceptance criteria: Meet the requirements

• UNIFORMITY OF DOSAGE UNITS (905):

Meet the requirements

IMPURITIES

Change to read:

• ORGANIC IMPURITIES

Buffer, Mobile phase, Dilute phosphoric acid, Diluent, Sample solution, and Chromatographic system: Proceed as directed in the *Assay*.

System suitability solution (*Procedure 1* may be used for Tablets containing sodium stearyl fumarate as an excipient in the formulation, and *Procedure 2* may be used for all products.)

Procedure 1: Place 1 Tablet of any potency into a vial, add 1 mL of water, and tightly seal the vial. Heat at 80° for about 30 h to generate a fumarate adduct of sitagliptin. Transfer the contents of the vial into a 100-mL volumetric flask, using a small amount of *Diluent*, and dilute with *Diluent* to volume. Mix well by stirring for 1 h. If needed, further dilute with *Diluent* to obtain a solution with a concentration of 0.08 mg/mL of sitagliptin, based on the label claim of the Tablet. Centrifuge a portion of the solution for 10 min or until the solution is clear, and use the supernatant.

Procedure 2: Place 10 mg of USP Sitagliptin Phosphate RS and 1 mg of sodium stearyl fumarate into a vial, add 1 mL of water, and tightly seal the vial. Heat at 80° for about 30 h to generate a fumarate adduct of sitagliptin. Transfer the contents of the vial into a 100-mL volumetric flask, using a small amount of *Diluent*, and dilute with *Diluent* to volume. Mix well by stirring for 1 h. Centrifuge a portion of the solution for 10 min or until the solution is clear, and use the supernatant.

Standard solution: 0.0002 mg/mL of USP Sitagliptin Phosphate RS in *Diluent*

Sensitivity solution: 0.0001 mg/mL of USP Sitagliptin Phosphate RS in *Diluent* from the *Standard solution*

System suitability

Samples: *System suitability solution* and *Sensitivity solution* [NOTE—See *Table 1* for relative retention times.]

Suitability requirements

Resolution: NLT 1.5 between sitagliptin and fumarate adduct, *System suitability solution*

Signal-to-noise ratio: NLT 10, *Sensitivity solution*

Analysis

Samples: *Sample solution* and *Standard solution*

Calculate the percentage of each impurity in the portion of Tablets taken:

$$\text{Result} = (r_U/r_S) \times (C_S/C_U) \times (1/F) \times (M_{r1}/M_{r2}) \times 100$$

- r_U

= peak response of each impurity from the *Sample solution*
- r_S

= peak response of sitagliptin from the *Standard solution*
- C_S

= concentration of USP Sitagliptin Phosphate RS in the *Standard solution* (mg/mL)
- C_U

= nominal concentration of sitagliptin in the *Sample solution* (mg/mL)
- F

= relative response factor (see *Table 1*)
- M_{r1}

= molecular weight of sitagliptin, 407.32
- M_{r2}

= molecular weight of anhydrous sitagliptin phosphate, 505.31

Acceptance criteria: See *Table 1*. ▲The reporting threshold is▲ (USP 1-Aug-2022) 0.1%.

Table 1

Name	Relative Retention Time	Relative Response Factor	Acceptance Criteria, NMT (%)
Sitagliptin acid ^a	0.55	0.65	0.2
Sitagliptin	1.0	—	—

Table 1 (continued)

Name	Relative Retention Time	Relative Response Factor	Acceptance Criteria, NMT (%)
Fumarate adduct (if present) ^{b, c}	1.2	1.0	0.2
Sitagliptin triazecine analog ^d	1.8	1.0	0.2
Sitagliptin styrylacetyl analog ^e	4.1	1.0	0.2
Sitagliptin phenylcrotonyl analog ^f	4.7	2.1	0.2
Any other individual impurity	—	1.0	0.2
Total impurities	—	—	0.6

^a (R)-3-Amino-4-(2,4,5-trifluorophenyl)butanoic acid.
^b 2-[[[(R)-4-Oxo-4-[3-(trifluoromethyl)-5,6-dihydro-[1,2,4]triazolo[4,3-a]pyrazin-7(8H)-yl]-1-(2,4,5-trifluorophenyl)butan-2-yl]amino)succinic acid.
^c This degradation product may be present only for Tablets containing sodium stearyl fumarate as an excipient in the formulation.
^d 10-(2,4,5-Trifluorobenzyl)-3-(trifluoromethyl)-6,7,10,11-tetrahydro-[1,2,4]triazolo[3,4-c][1,4,7]triazecine-8,12(5H,9H)-dione.
^e (E)-1-[3-(Trifluoromethyl)-5,6-dihydro-[1,2,4]triazolo[4,3-a]pyrazin-7(8H)-yl]-4-(2,4,5-trifluorophenyl)but-3-en-1-one.
^f (E)-1-[3-(Trifluoromethyl)-5,6-dihydro-[1,2,4]triazolo[4,3-a]pyrazin-7(8H)-yl]-4-(2,4,5-trifluorophenyl)but-2-en-1-one.

ADDITIONAL REQUIREMENTS

- **PACKAGING AND STORAGE:** Preserve in well-closed containers. Store at controlled room temperature.
- **USP REFERENCE STANDARDS** <11>
USP Sitagliptin Phosphate RS